

Solving equations

Undoing what was done to the unknown — and checking the root in the equation you were given.

LESSON 16–18

3 × 40 MIN

MATEMATIKA 6, §4

S7 2.5

LESSON CLOCK

20'

25'

30'

30'

15'

The balance	20 min
One and two steps	25 min
Brackets and both sides	30 min
Word problems	30 min
Homework	15 min

LEARNING OBJECTIVES

- Solve one-step and two-step equations by inverse operations.
- Solve equations with a bracket and with the unknown on both sides.
- Check a root by substituting into the original equation.
- Write a simple word problem as an equation.

TEXTBOOK

National	pp. 43–50
Cambridge	Stage 7, pp. 32–38
Workbook	Exercise 2.5

01

Explanation

The balance

An equation is a balance: whatever is done to one side must be done to the other, or it tips.

OPERATION ON THE UNKNOWN	UNDONE BY
adding 5	subtracting 5
subtracting 5	adding 5
multiplying by 3	dividing by 3
dividing by 3	multiplying by 3

UNDO IN THE REVERSE ORDER

In $3x + 5$ the x was multiplied first and added to second, so the addition is undone first and the multiplication second — like taking off shoes before socks were put on.

One and two steps

EQUATION	STEP	ROOT
$x + 7 = 12$	subtract 7	$x = 5$
$x - 4 = 9$	add 4	$x = 13$
$5x = 40$	divide by 5	$x = 8$
$\frac{x}{3} = 6$	multiply by 3	$x = 18$
$3x + 5 = 20$	subtract 5, divide by 3	$x = 5$
$2x - 7 = 11$	add 7, divide by 2	$x = 9$

CHECK IN THE EQUATION AS IT WAS GIVEN

For $3x + 5 = 20$: $3 \cdot 5 + 5 = 20 \checkmark$. Checking in a line of your own working repeats any error you have already made.

Brackets and both sides

EQUATION	FIRST MOVE	ROOT
$2(x + 3) = 14$	expand or divide by 2	$x = 4$
$3(x - 1) = 12$	divide by 3	$x = 5$
$5x = 2x + 12$	subtract $2x$	$x = 4$
$4x + 1 = 2x + 9$	subtract $2x$, then 1	$x = 4$

GATHER THE UNKNOWNNS ON THE SIDE WHERE MORE OF THEM ARE

In $4x + 1 = 2x + 9$ taking $2x$ from both sides leaves $2x$ on the left, which is positive. Working the other way gives $-2x$ and an extra chance to slip.

Word problems

PROBLEM	EQUATION	ANSWER
a number trebled and increased by 5 is 26	$3x + 5 = 26$	7
a rectangle of perimeter 30 with width 6	$2(6 + b) = 30$	$b = 9$
5 pens cost 20 000	$5p = 20\,000$	4000
Ali is 4 years older than Sardor; together 26	$x + (x + 4) = 26$	11 and 15

THREE LINES: LET, EQUATION, ANSWER

Say what the letter stands for, write the equation, solve it, and answer the question in words. The first and last lines carry as many marks as the algebra between them.

02 Worked examples

EXAMPLE 1 Solve $3x + 5 = 20$.

- 1 Subtract 5 from both sides: $3x = 15$.
- 2 Divide both sides by 3: $x = 5$.
- 3 Check: $3 \cdot 5 + 5 = 20$ ✓

ANSWER

$x = 5$

EXAMPLE 2 Solve $4x + 1 = 2x + 9$.

① Subtract $2x$: $2x + 1 = 9$.

② Subtract 1 : $2x = 8$.

③ Divide by 2 : $x = 4$.

④ Check: $17 = 17$ ✓

ANSWER

$$x = 4$$

EXAMPLE 3 A rectangle has perimeter 30 cm and width 6 cm . Find its length.

① Let the length be b .

② $2(6 + b) = 30$

③ $6 + b = 15$

Divide by 2 .

④ $b = 9\text{ cm}$

ANSWER

$$9\text{ cm}$$

03 Interactive model

Draw a pair of scales on the board and physically remove the same weight from both pans; every later step of algebra is that picture.

One step at a time

$x + 7 = 12$ gives:

$x = 19$

$x = 5$

$x = 7$

$x = 12$

$5x = 40$ gives:

$x = 35$

$x = 45$

$x = 8$

$x = 200$

$\frac{x}{3} = 6$ gives:

$x = 2$

$x = 18$

$x = 9$

$x = 3$

In $3x + 5 = 20$ you first:

divide by 3

subtract 5

add 5

multiply by 3

It gives:

$x = 5$

$x = 15$

$x = 8$

$x = 25$

$2(x + 3) = 14$ gives:

$x = 4$

$x = 7$

$x = 11$

$x = 8$

$5x = 2x + 12$ gives:

$x = 4$

$x = 12$

$x = 3$

$x = 6$

A root should be checked in:

your last line

the original equation

a calculator

nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Equation	Tenglama	Уравнение
Root	Ildiz	Корень
Unknown	Noma'lum	Неизвестное
Inverse operation	Teskari amal	Обратное действие
To balance	Muvozanatlash	Уравновешивать
Both sides	Ikkala tomon	Обе части
To check	Tekshirmoq	Проверить
Solution	Yechim	Решение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An equation is like:

a list

a balance

a graph

a table

Multiplying by 3 is undone by:

adding 3

subtracting 3

dividing by 3

squaring

$2x - 7 = 11$ gives:

$x = 2$

$x = 9$

$x = 18$

$x = 4$

$3(x - 1) = 12$ gives:

In $4x + 1 = 2x + 9$ you gather the unknowns:

A word problem needs first:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x + 7 = 12$

ANSWER $x = 5$

2. $x - 4 = 9$

ANSWER $x = 13$

3. $5x = 40$

ANSWER $x = 8$

4. $\frac{x}{3} = 6$

ANSWER $x = 18$

5. $3x + 5 = 20$

ANSWER $x = 5$

6. $2x - 7 = 11$

ANSWER $x = 9$

7. $x + 12 = 12$

ANSWER $x = 0$ **Medium · the standard question**

7 PROBLEMS

1. $2(x + 3) = 14$

ANSWER $x = 4$

2. $3(x - 1) = 12$

ANSWER $x = 5$

3. $5x = 2x + 12$

ANSWER $x = 4$

4. $4x + 1 = 2x + 9$

ANSWER $x = 4$

5. A number trebled and increased by 5 is 26

ANSWER 7

6. 5 pens cost 20 000: one pen

ANSWER 4000

7. A rectangle of perimeter 30 and width 6

ANSWER Length 9

Hard · stretch and reason

7 PROBLEMS

1. $4(x - 2) = 2(x + 3)$

ANSWER $x = 7$

2. $5x - 3 = 3x + 7$

ANSWER $x = 5$

3. $\frac{x}{2} + 4 = 10$

ANSWER $x = 12$

4. $7 - x = 2$

ANSWER $x = 5$

5. Ali is 4 years older than Sardor and together they are 26

ANSWER 11 and 15

6. Three consecutive numbers add to 42

ANSWER 13, 14, 15

7. Why check in the original equation?

ANSWER A check on your own working repeats your own error

07 Homework**Homework — 5 tasks**

Write the check as its own line beneath every answer.

1. Solve $x + 9 = 21$, $6x = 54$ and $\frac{x}{4} = 7$.

2. Solve $4x + 3 = 27$ and $5x - 8 = 12$.

3. Solve $3(x + 2) = 21$.

4. Solve $6x + 2 = 4x + 14$.

5. A number doubled and increased by 7 is 31. Find it, writing all three lines.